



Cancer screening

saving lives or diagnosing more sickness?
and the resulting effect on the cost of insurance cover

Adèle Groyer

Health Claims Forum

5 December 2013

Say aaaah





> In spitting distance



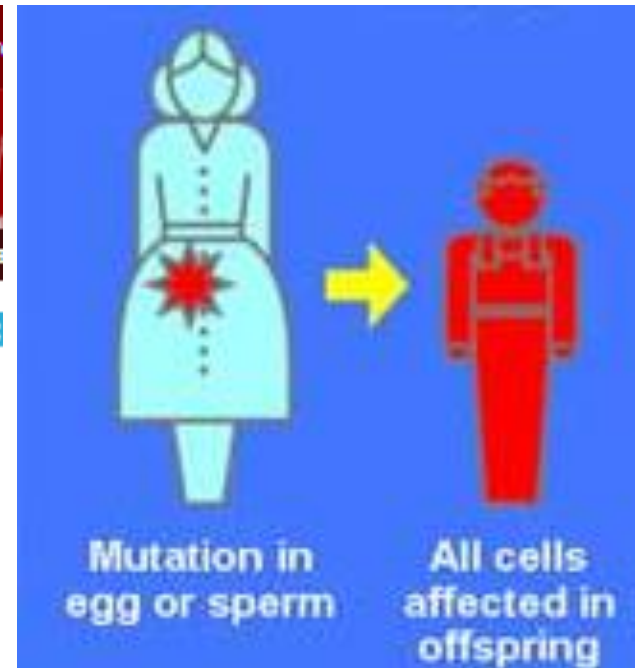
Breakthrough? Scientists unveil the 'single biggest leap forward' in tackling prostate cancer

Men could receive routine saliva test within five years, after pan-European research

STEVE CONNOR  WEDNESDAY 27 MARCH 2013



- > In spitting distance
- > Spitting image





- > In spitting distance
- > Spitting image





> In spitting distance

> Spitting image

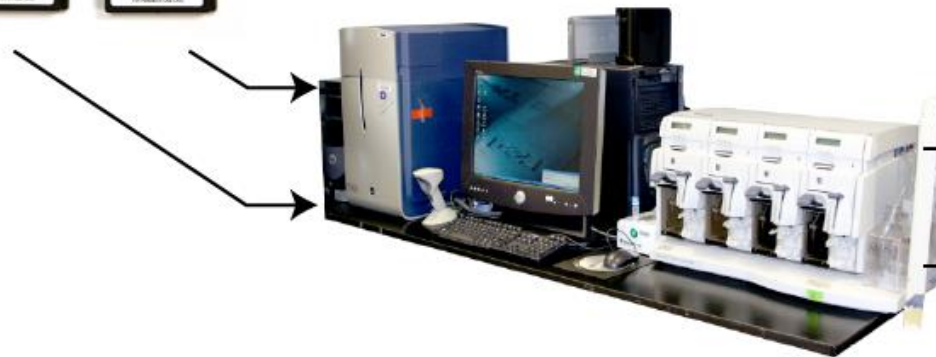
Prepare RNA from

"Normal"

Tumor



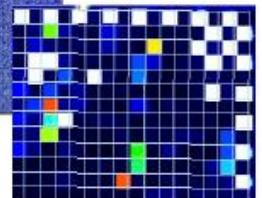
Label with Biotin



Tumor



Normal





- > In spitting distance
- > Spitting image
- > Spitting mad



The Telegraph

HOME NEWS WORLD SPORT **FINANCE** COMMENT BLOGS CULTURE TRAVEL LIFE
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Could new cancer tests make you uninsurable?

A £5 test could soon allow doctors to see how likely you are to develop cancer, this could make a huge difference to the cost of life insurance or critical illness cover.

GENETIC TESTING MORATORIUM???

Is cancer genetic?



MAJORITY Acquired

Somatic mutations

- Occur in **nongermline** tissues
- Are **nonheritable**



Nonheritable

Somatic mutation
(e.g., breast)

MINORITY Inherited

Germline mutations

- Present in egg or sperm
- Are **heritable**
- Cause cancer family syndrome



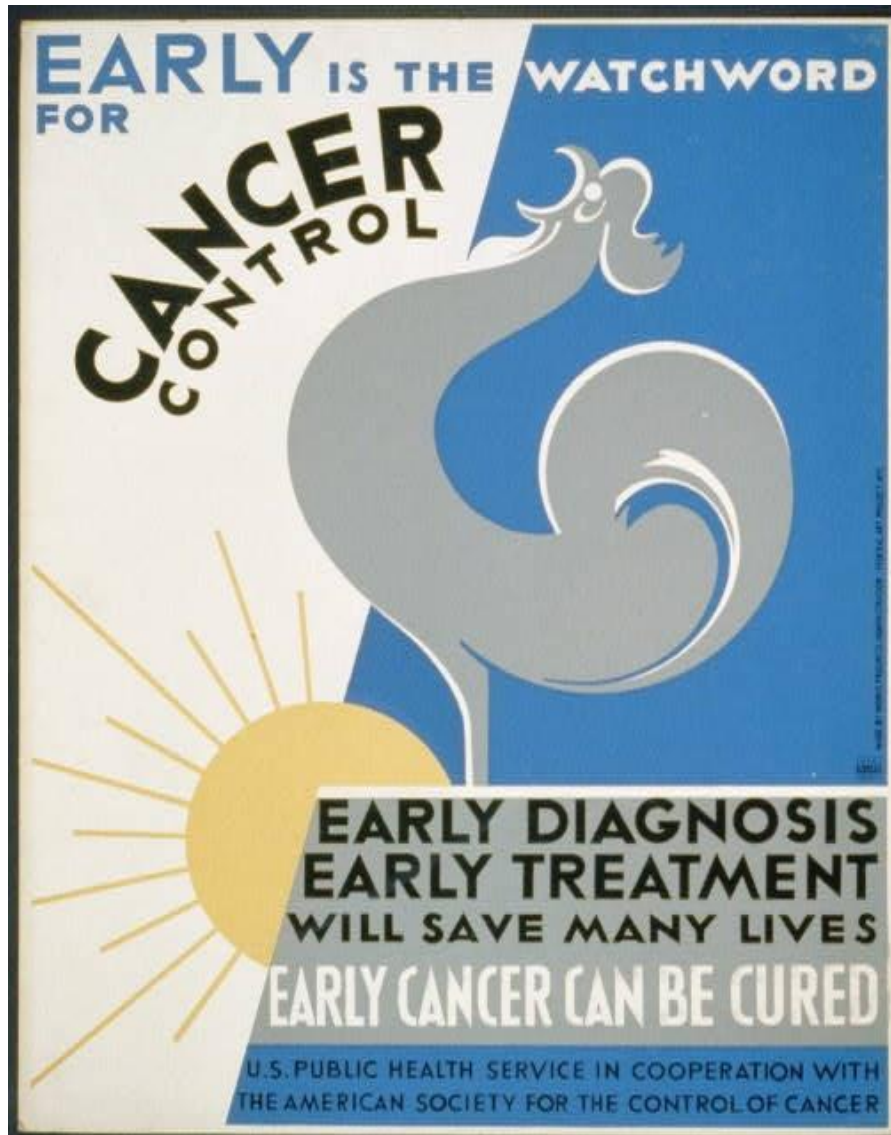
Mutation in
egg or sperm

All cells
affected in
offspring

Adapted by Joanne Kelly, © 2004

Source: www.cancer.gov/cancertopics/understandingcancer/cancergenomics/

Is early detection always a good thing?



OVER-DIAGNOSED

MAKING PEOPLE SICK IN
THE PURSUIT OF HEALTH

DR. H. GILBERT WELCH,
DR. LISA M. SCHWARTZ, AND DR. STEVEN WOLOSHIN

"This brilliantly researched, well-argued, and clearly written book will help us avoid the unnecessary tests, drugs, surgeries, and anxiety that are the inevitable outcomes of our epidemic of overdiagnosis."

—SIDNEY WOLFE, MD, author of
Worst Pills, Best Pills and editor of *WorstPills.org*





- > Screening available now – effect on cancer incidence
- > How screening programmes are evaluated
- > Screening in development
- > Implications for life and health insurance



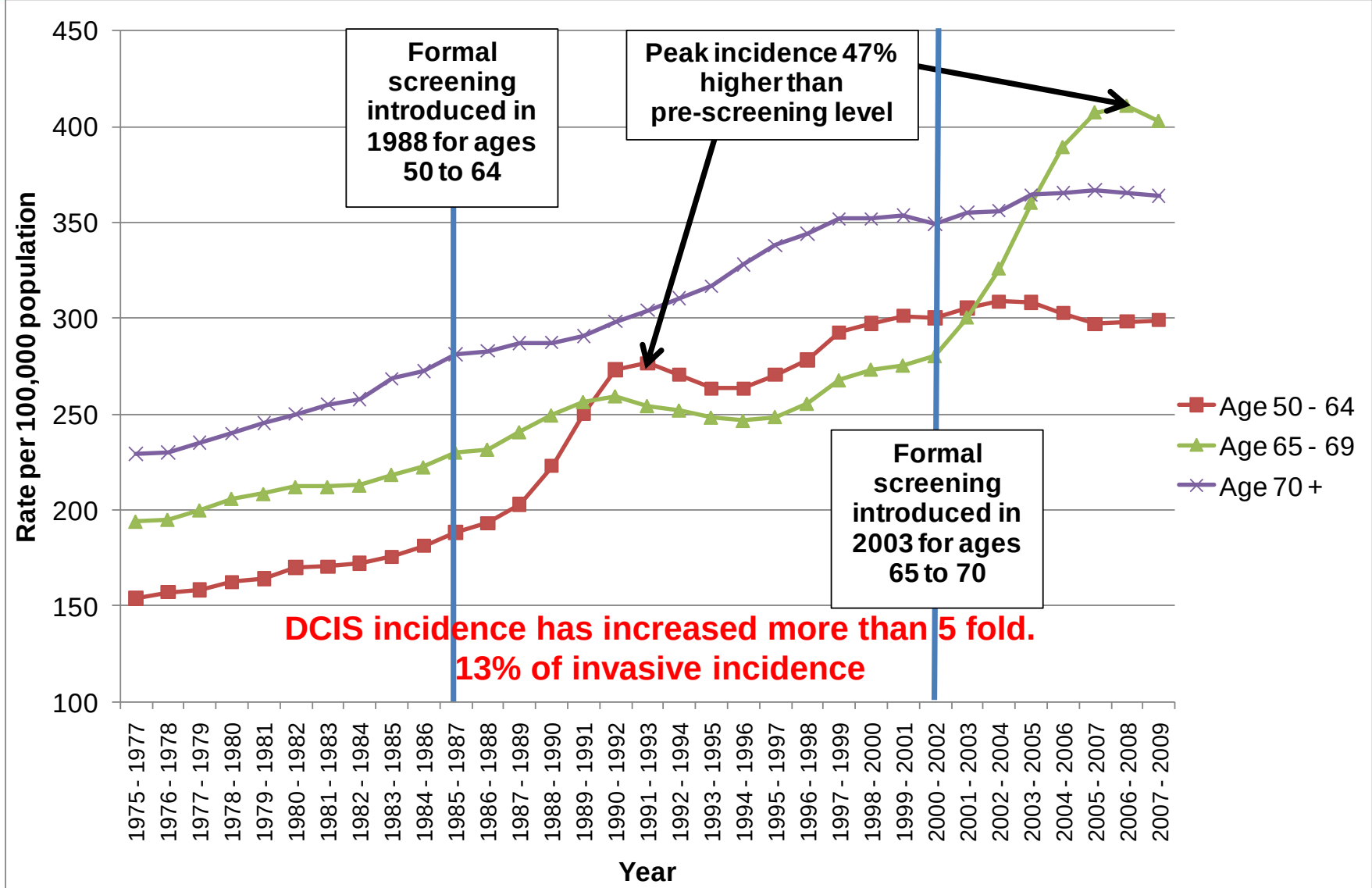
Effect of screening on cancer incidence

Formal screening programmes

Breast, Cervix, Bowel



Invasive breast cancer incidence in England





Effect of screening on cancer incidence

Informal screening programmes



Prostate cancer

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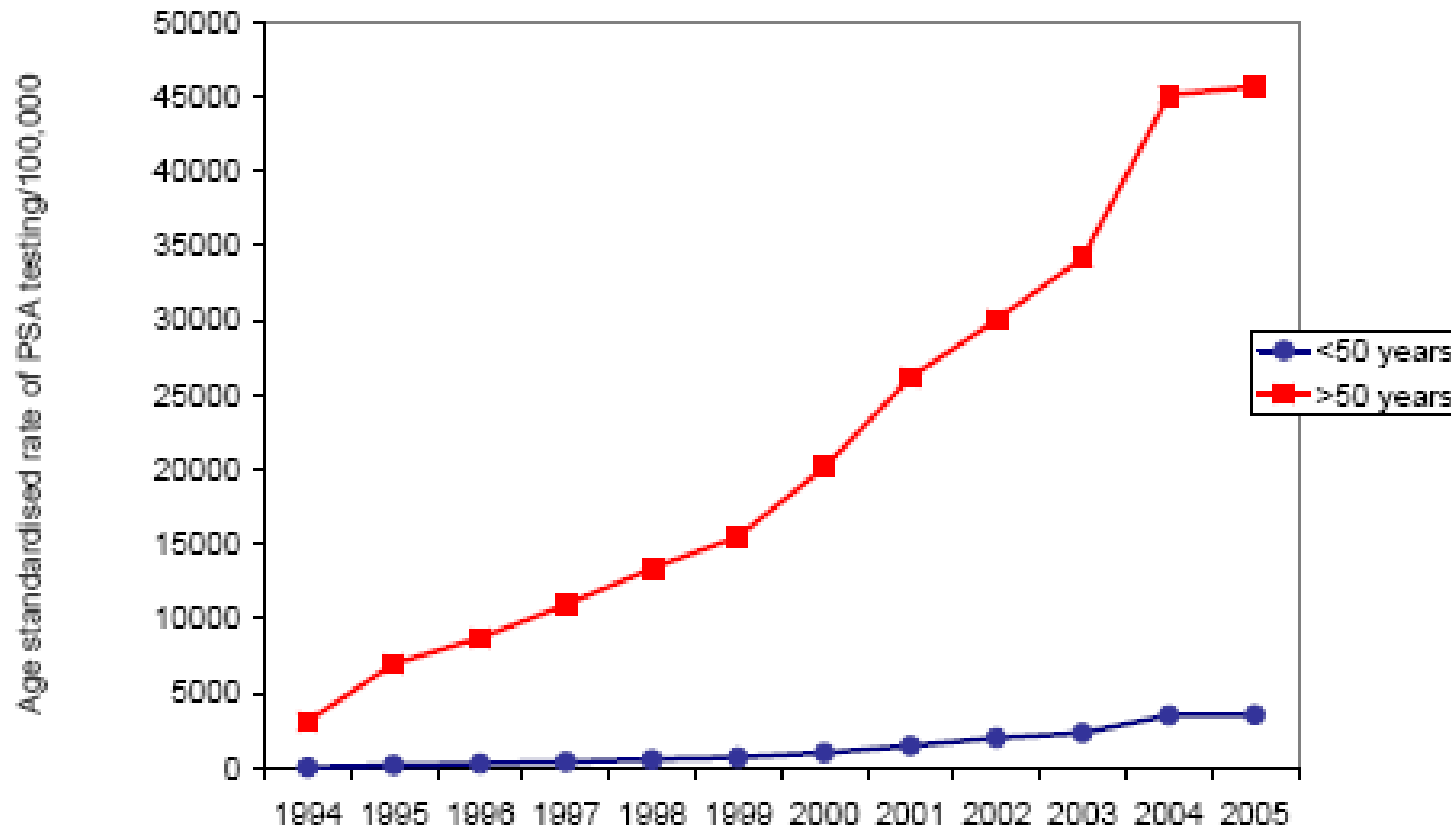
"It may be more inconvenient, but the 'Reverse Prostate Exam' is a lot less embarrassing for the both of us."

search ID: mbcn1967

Prostate cancer screening in Ireland



Figure 5. Annual number of PSA tests 1994-2005



16% of Irish men had a PSA test in 2004.

In the UK the figure was 8.6%.

Source: "Recent trends in Prostate Cancer"

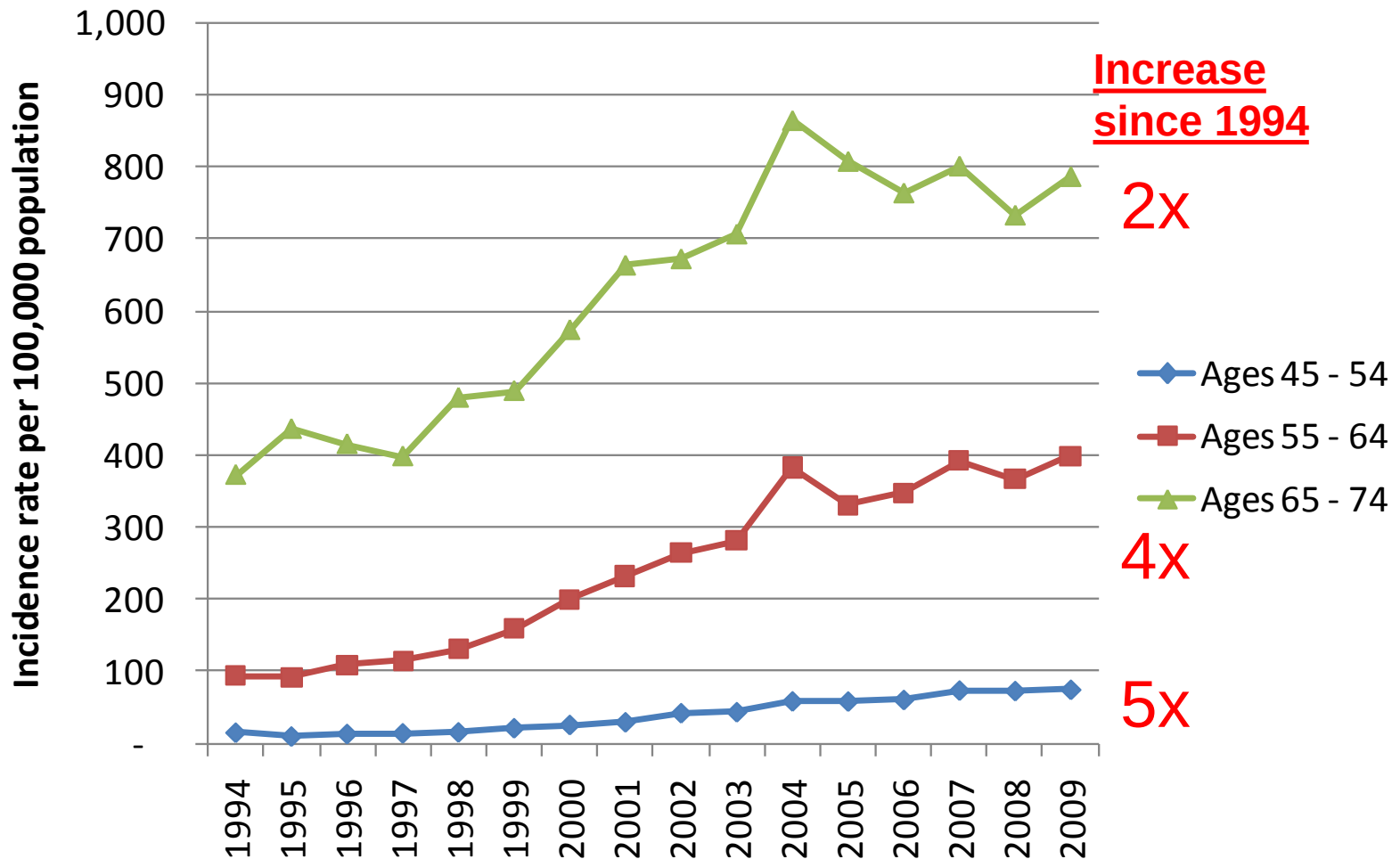
National Cancer Registry Ireland publication Cancer trends No. 3

Figure 6. Prostate cancer mortality, Ireland 1950-2006

Prostate cancer incidence in Ireland



Trends in prostate cancer incidence in Ireland

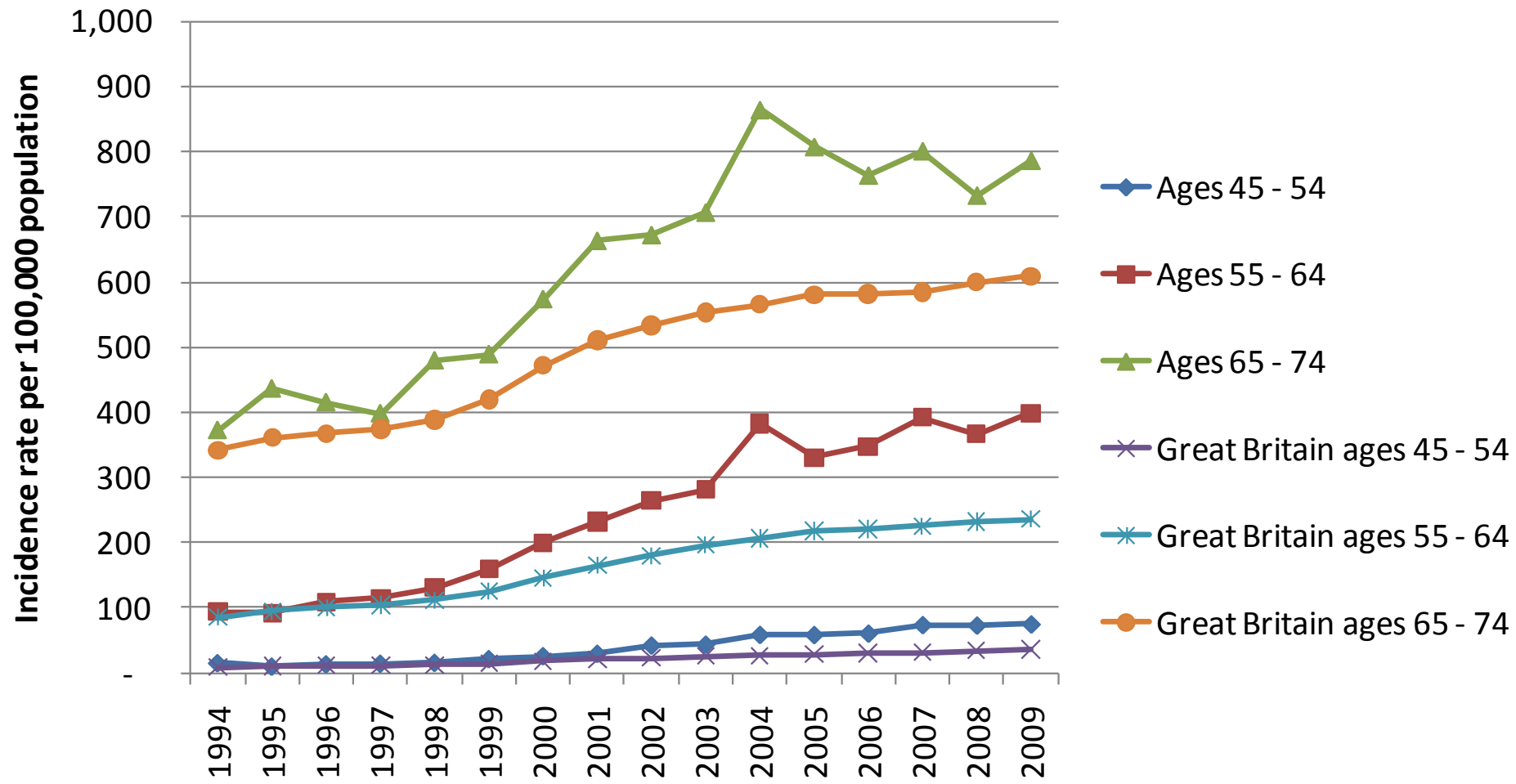


Source: National Cancer Registry Ireland Incidence Data

Double the screening, double the incidence

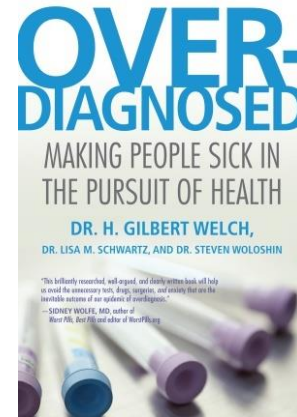
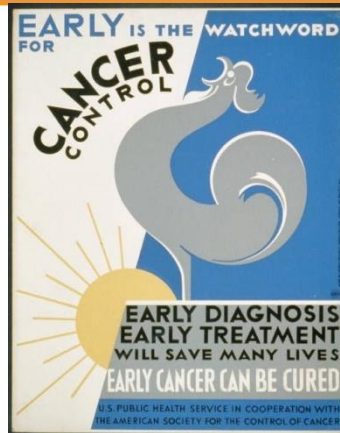


Trends in prostate cancer incidence in Ireland



Sources: National Cancer Registry Ireland Incidence Data and Cancer Research UK website

Diagnosing a lot of prostate cancer to save lives



Country	Age standardised incidence rate per 100,000	Incidence rank out of 184 countries	Age standardised mortality rate per 100,000
Ireland	126.3*	3	13.0
U.S.A.	83.8	18	9.8
U.K.	64.0	33	13.8

* Now equal to rate in USA 1995 - 2005

Source: Ferlay J, Shin HR, Bray F, Forman D, Mathers C and Parkin DM.

GLOBOCAN 2008 v2.0, Cancer Incidence and Mortality Worldwide: IARC CancerBase No. 10 [Internet]. Lyon, France: International Agency for Research on Cancer; 2010. Available from: <http://globocan.iarc.fr>, accessed on 07/03/2013.

Reservoir of undetected cancer



UK cancer prevalence estimated at 3%, 4% ages 45 – 64, 13% ages 65+

J. Maddams et al. Cancer prevalence in the United Kingdom: estimates for 2008, Br J Cancer. 2009 August 4; 101(3): 541–547

Cancer site	Age range	Autopsy surprise prevalence	Source
Prostate cancer	All ages 50 - 59 60 – 69	12% approx 25% approx 33%	Yin M et al, <i>J Urol</i> . 2008 Mar;179(3):892-5 Data from 1994 – 2007 (PSA era)
Thyroid cancer	All	35.6%	Harach HR et al, <i>Cancer</i> . 1985 Aug 1;56(3):531-8.
Invasive breast cancer	All 45 - 54	0% to 1.8% 7%	Welch H and Black W <i>Ann Intern Med</i> . 1 December 1997;127(11):1023-1028
DCIS	All 40 - 70	0% to 14.7% 0% to 39%	Welch H and Black W <i>Ann Intern Med</i> . 1 December 1997;127(11):1023-1028
Lung cancer	All <70	0.7% (15% increase) 0.8%	Chan CK et al, <i>Chest</i> 1989; 96:291-96 1973 – 1982 data used
Uterine cancer	All	4 – 6x population incidence	Horwitz RI et al, <i>Lancet</i> . 1981 Jul 11; 2(8237):66-8.



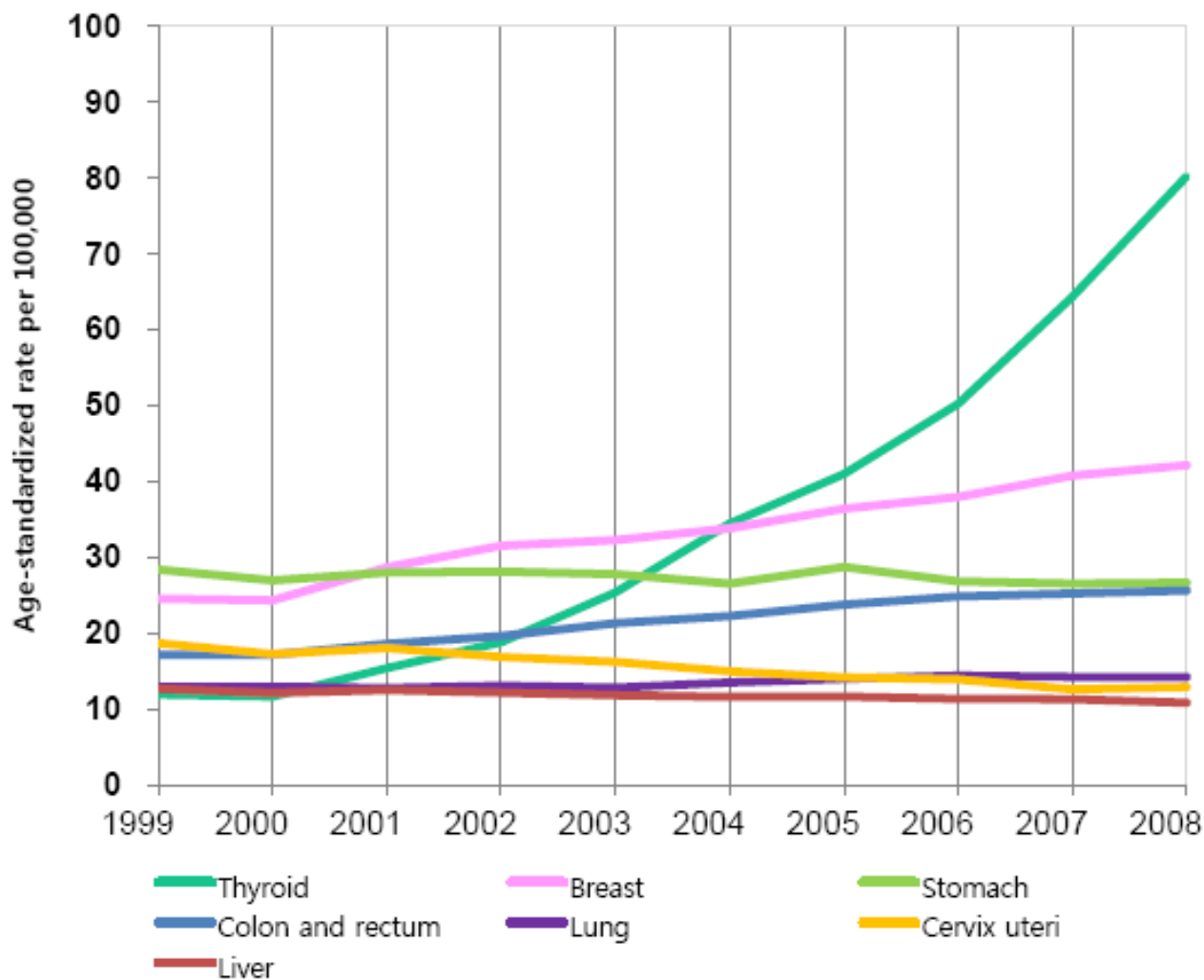
Thyroid cancer



In 2009 13.2% of South Korean adults said they had undergone screening by thyroid ultrasonography at some stage.

Asian Pacific J Cancer Prev, 12, 1657-1663

Thyroid cancer incidence in Korea (females)



Site	Year		Annual Percent Change (%)
	1999	2008	
Thyroid	11.9	80.2	25.7*
Breast	24.5	42.1	6.5*
Stomach	28.3	26.6	-0.5
Colon and rectum	17.1	25.6	5.2*
Lung	12.9	14.2	1.5*
Cervix uteri	18.6	12.9	-4.4*
Liver	12.6	10.8	-1.5*

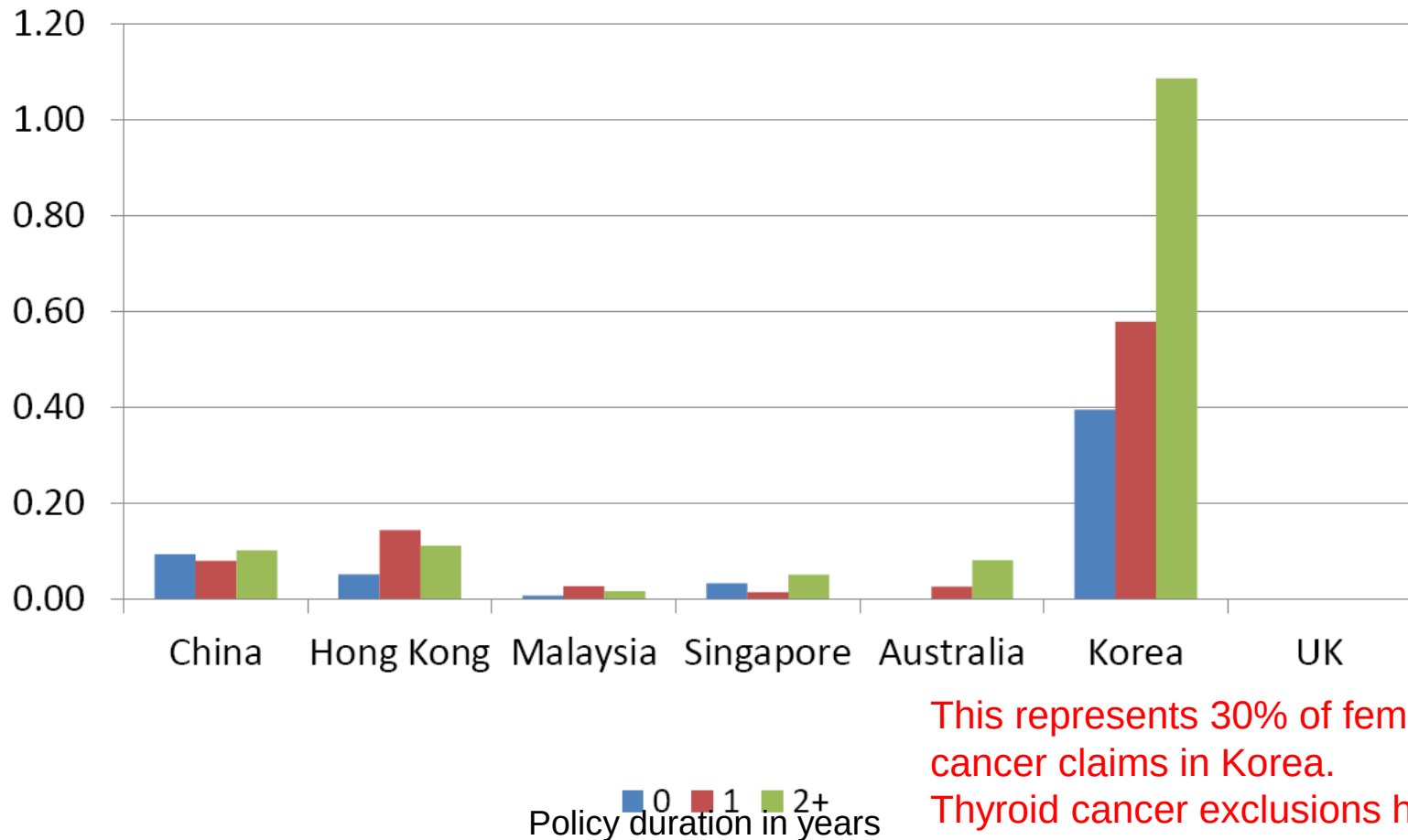
* P < .05

Age-standardized incidence Rate(ASR) uses "mid-year population in 2000" as standard population.

Source: National Cancer Registration & Statistics 2008, the Korea Central Cancer Registry
Health Claims Forum | 5 December 2013



Female Thyroid Cancer Incidence per 1,000



This represents 30% of female cancer claims in Korea.
Thyroid cancer exclusions have been introduced.

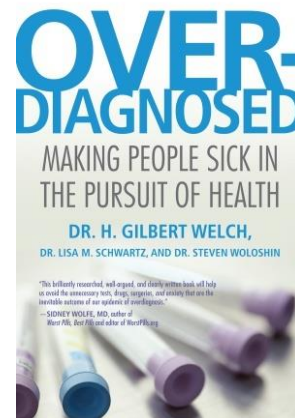
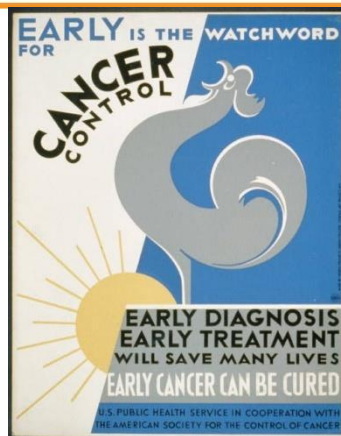
May 2012

Gen Re Dread Disease Survey 2004-200
Initial Results

Health Claims Forum | 5 December 2013

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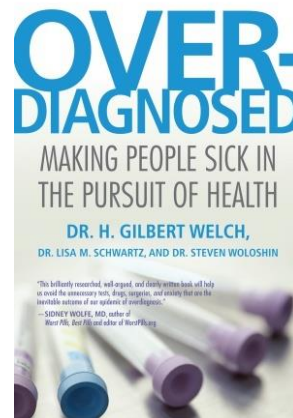
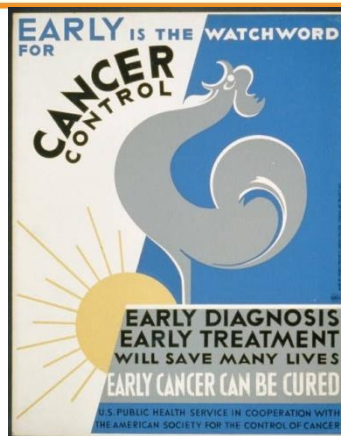
Diagnosing sickness – thyroid cancer



Country	World age-standardised incidence rate per 100k	Age standardised mortality rate per 100k	Annual change since 1998
Republic of Korea	35.44	0.53	+25.7%
U.S.A.	9.90	0.28	+6.5% SEER 13 age-adjusted incidence trends 1998-2009
U.K.	2.78	0.26	+5% Cancer Research UK EASR trend

Source: Ferlay J, Shin HR, Bray F, Forman D, Mathers C and Parkin DM. GLOBOCAN 2008 v2.0, Cancer Incidence and Mortality Worldwide: IARC CancerBase No. 10 [Internet]. Lyon, France: International Agency for Research on Cancer; 2010. Available from: <http://globocan.iarc.fr>, accessed on 07/03/2013.

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When is screening justified?

What policy makers think about

It's a benefit-harm-cost equation



Benefits

**Lower
mortality**

**Lower
morbidity**

Harms

Treatment

*physical & psychological
harms*

Screening & Diagnostic procedures

*e.g. CT-scan radiation,
colonoscopy complications*

Screening psychology

*worry awaiting results or false +ve
OR false security from false -ve*

Costs

Programme resources

clinical & admin

Treatment

for over-diagnosed cases

**Less drastic
treatment required**
early diagnosis

**Opportunity cost for
addressing other
health problems**



“The only way to know if the screening is saving lives is by doing a randomized trial. It's easy to forget this and assume that if technology can find more cancer, it will save more lives. Marketers exploit this assumption. Don't fall for it.”

— H. Gilbert Welch, *Overdiagnosed: Making People Sick in the Pursuit of Health*



Screening trial statistics



	Breast cancer	Bowel cancer	Prostate cancer
Screening method	Mammography	Faecal Occult Blood Test	PSA testing every 4 yrs
Source	Duffy, S et al; J Med Screen. 2010 March Swedish Two-County trial and England pre&post-screening introduction <i>Contentious data sets contaminated / age ranges considered</i>	Robinson, M et al; Gut 1999;45:588-592 Nottingham randomised controlled trial of faecal occult blood screening	Schröder, F et al; N Engl J Med 2009;360:1320-8. European Randomized Study of Screening for Prostate Cancer
Age range	50 - 69	45 - 74	50 - 74
Screen vs control group	Inferred because of lack of clear control group	15% lower bowel cancer mortality	20% lower prostate cancer mortality
Number needed to screen to save 1 life	114 - 175 over 20 years	747 over 7.8 years	1,410 over 9 years
Number needed to treat to save 1 life	1.4 to 1.5 over 20 years	3.9 (at most)	48 over 9 years
Number harmed by screening to save 1 life		0.1	NOT ADVOCATED = NOT SURPRISING!!!



UNDER INVESTIGATION

freedigitalphotos.net

Prostate, Lung, Colorectal and Ovarian (PLCO) Cancer Screening Trial



- > Population based randomised trial in the United States
- > To determine the effects of screening on cancer-related mortality and secondary endpoints in men and women aged 55 to 74
- > Over 77,000 participants
- > The screening component of the trial was conducted between 1993 and 2006.
- > Participants are being followed until 2015.



> UK Collaborative Trial of Ovarian Cancer Screening

200,000 subjects
recruited from
population registers

Post-menopausal women
aged 50 – 74 recruited
between 2001 and 2005

No screening
(control): **100,000**

Annual ultrasound:
50,000

Blood test for Serum
CA125 with ultrasound
for recall: **50,000**

- > Compare mortality rates in the screening arms after 7-year follow-up
- > Annual screening phase is now complete
- > Results to be reported in 2015

Source: http://www.instituteforwomenshealth.ucl.ac.uk/academic_research/gynaecologicalcancer/gcrc/ukctocs

UKCTOCS Ovarian cancer trial prevalence screening round results



Category	Study arm	Ultrasound only	Blood test + ultrasound
	Undergoing first screen	48,230	50,078
Total +ve	Surgery biopsy via laparotomy / laparoscopy	845	97
TRUE +ve	Cancers detected by screening	45 (24 invasive)	42 (34 invasive)
FALSE +ve		800	45
FALSE -ve	Symptomatically diagnosed cancer within 1 year	8	5
TRUE -ve		47,377	49,976
	Sensitivity (true +ve) / (true +ve + false -ve)	84.9%	89.4%
	Specificity (true -ve) / (true -ve + false +ve)	98.3%	99.9%
	Positive predictive value true positives as a % of test positives (i.e. cancers as a % of all surgeries)	5.3%	43.3%

Source: Menon U et al, Sensitivity and specificity of multimodal and ultrasound screening for ovarian cancer, and stage distribution of detected cancers: results of the prevalence screen of the UK Collaborative Trial of Ovarian Cancer Screening (UKCTOCS), [Lancet Oncol.](#) 2009 Apr;10(4):327-40

Lung cancer screening trials



Lpa

United States National Lung Screening Trial

- Randomised study of 53,000 heavy (ex) smokers **ended in 2010**.
- Results showed **20% lower mortality in group screened by CT-scan vs chest X-ray**.
- Suggests benefit to diagnosing lung cancer early.

UK Lung Cancer Screening Trial

- Randomised controlled trial pursued in light of NSLT findings **started in 2012**.
- In pilot phase with 4,000 general population subjects.
- **Questionnaire sent to all with high risk cases in screening arm recalled for CT-scan**.
- Larger 32,000 subject, 10-year study to compare mortality between study arms.

Scotland Early CDT –Lung blood test trial

- Randomised study of 10,000 heavy (ex) smokers **started in 2012**.
- **Blood test** to identify antibodies generated as an immune response to lung cancer.
- **CT-scan follow up** for positives. Fewer false positives are generated vs CT-scan only.
- Study to compare clinical outcomes and cost of care between screening arm **(blood test + CT-scan) vs (chest X-rays only)**



EarlyCDT -Lung *Find lung cancer at its earliest stages.*



What is it?

Blood test for circulating antibodies to tumour proteins

CAGE, GBU4-5, HuD, MAGE, A4, NY-ESO-1, p53, SOX-2

Cost: £210 to £240



Sensitivity: 40%



Specificity: 90%



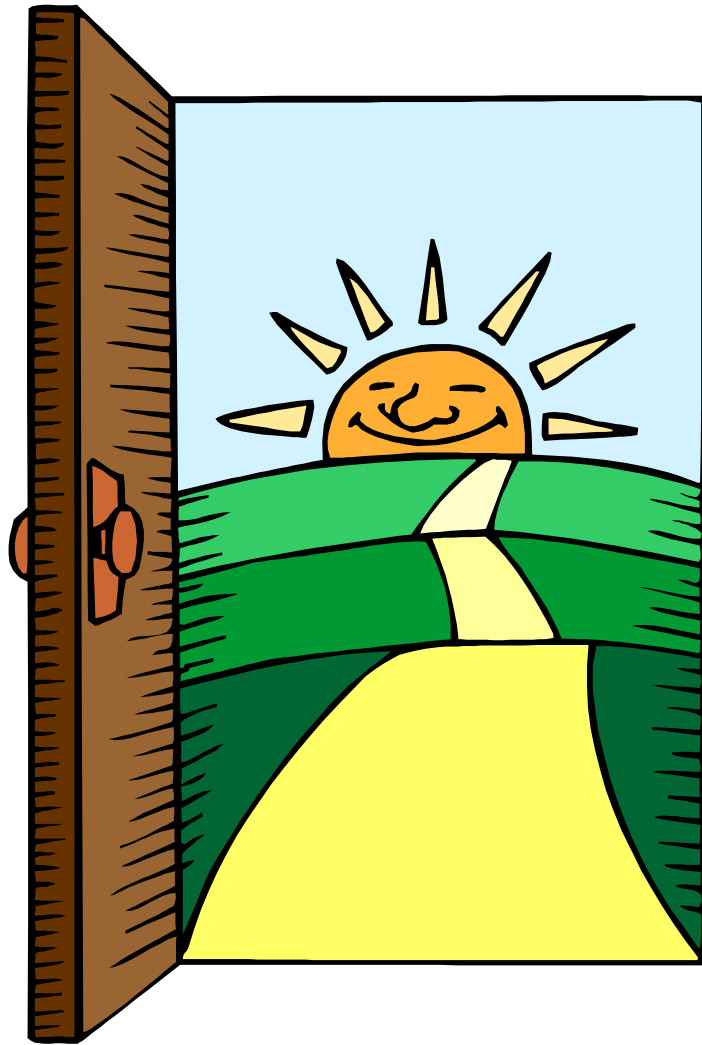
PPV: 10%

Follow-up with CT-scan



Implications

- > 5-year age shift could double incidence rates for people in their 50s.
- > Oncimmune aims to use this technology to screen for 90% of solid cancers.



Other developments

Treatment pathways



- > UK Study to find the best way of treating localised prostate cancer
 - *Radiotherapy*
 - *Prostatectomy*
 - *Active monitoring of PSA levels*
- > Over 100,000 participants recruited between 2001 and 2008
- > Now in follow-up phase
- > **Implication for screening:**
If active monitoring is found to be effective, over-diagnosis is potentially less harmful



Genomic tests and treatment decisions



Have you been
newly diagnosed with early-stage breast cancer?

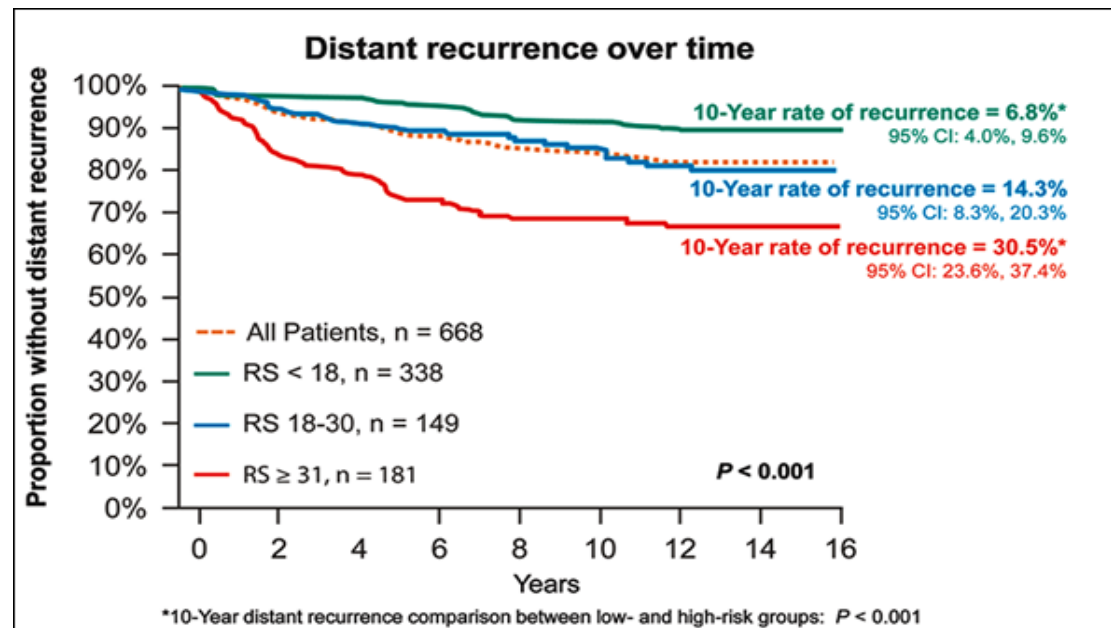
Have you discussed
whether chemotherapy will be part
of your treatment plan?



oncotype DX[®]
Breast Cancer Assay

This guide is designed to educate newly diagnosed women with early-stage breast cancer about Oncotype DX[®], a diagnostic test that may help you and your doctor make a more informed treatment decision.

- > Gene expression analysis on cancer tissue
- > Predicts likelihood recurrence based on expression of 21 genes
- > Predicts benefits of adjuvant therapy



RS = recurrence score

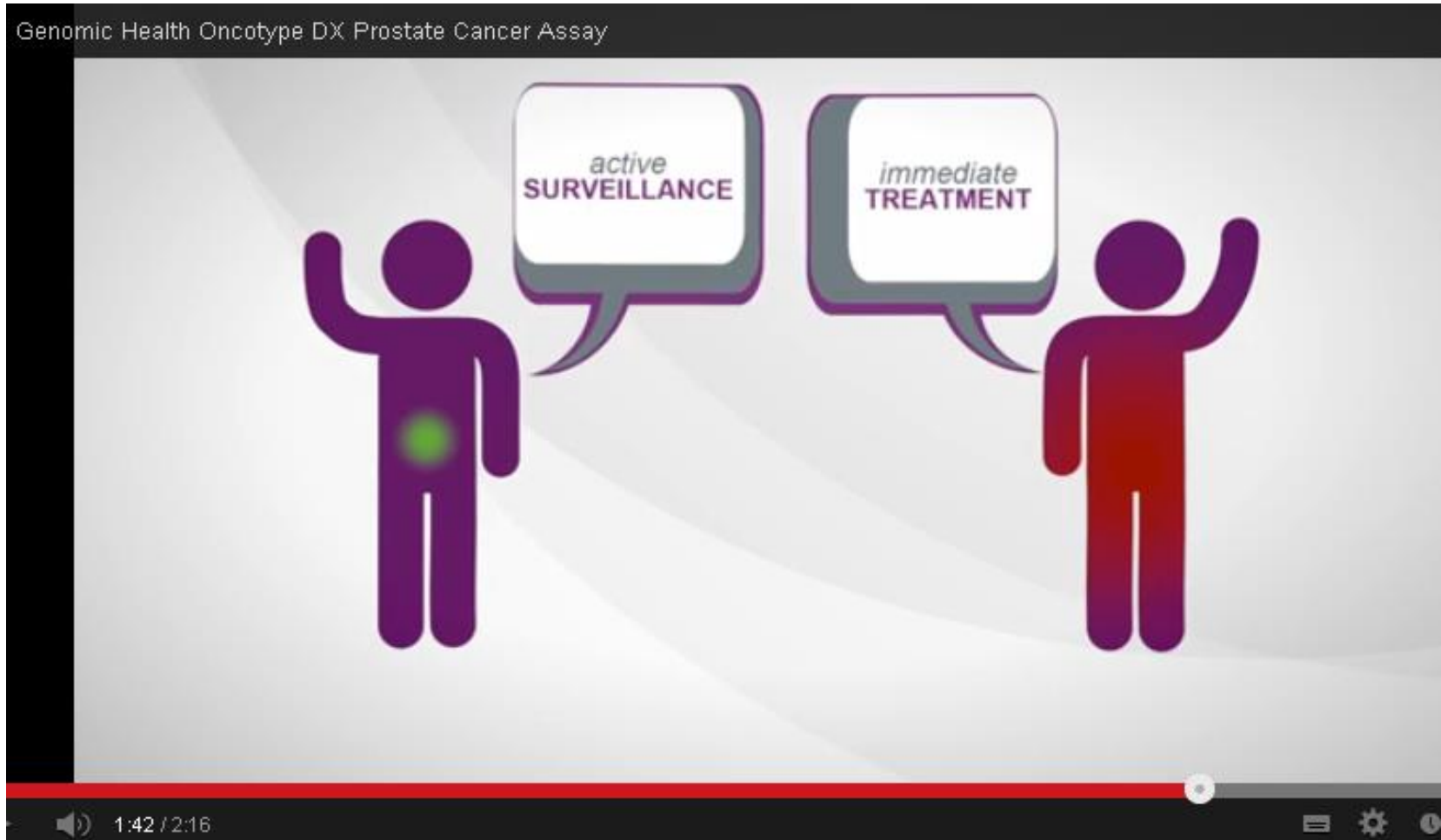
for colon and prostate cancers too



oncotype DX[®]
Breast Cancer Assay

oncotype DX[®]
Colon Cancer Assay

oncotype DX[®]
Prostate Cancer Assay

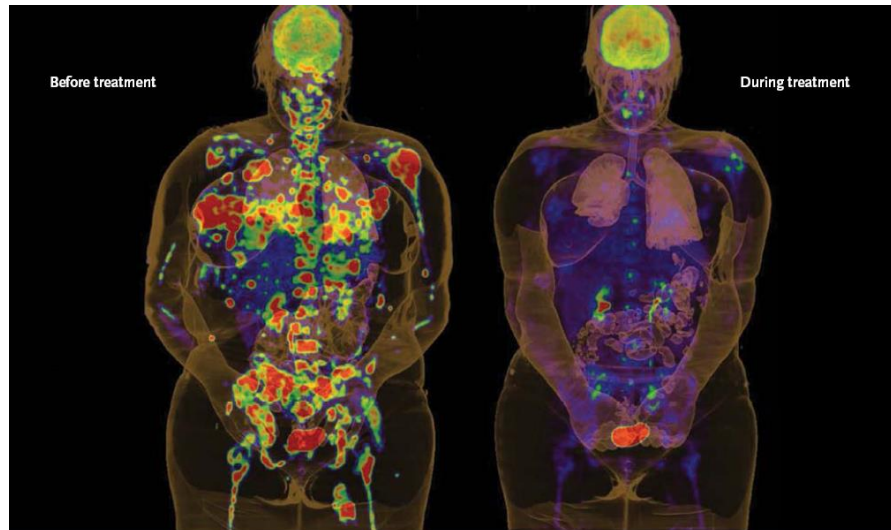


Source: <http://prostate-cancer.oncotypedx.com/prostate/patient.aspx>



PHARMACEUTICAL COMPANIES OWN GENETIC TESTS AND DEVELOP THE MEDICINES THAT GO WITH THE RESULTS

> Melanoma – BRAF Gene mutation – Vemurafenib



- > Chronic Myeloid Leukaemia – Imatinib/Gleevec
- > Acute Lymphatic Leukaemia – DNA testing routine
- > Gastro-Intestinal Stromal Tumours GISTs – Imatinib



What should (re)insurers be thinking about?





>Critical Illness

- Dangers of diagnosis-based definitions
 - are **new exclusions** needed?
 - are **severity measures** needed?

vs

- **Opportunities to differentiate** by **extending cover** to include radical treatments for screen-detected in situ-cancers or preventative treatments informed by genetic pre-disposition
- **Partial / tiered payments** to provide extensive cover at reduced cost





> Life cover **COSTS DOWN**

- > Benefits of earlier diagnosis
- > Targeted treatment

> CI cover **COSTS UP**

- Increase in diagnosis rates
- Further costs arising from expansion of cover to screen-detected early cancers

> Income protection: **UNCLEAR**

- Inceptions
 - could increase with more diagnoses
 - or decrease if treatment is quick / mild
 - increase more likely on short deferred periods
- Terminations
 - could increase with quick cures
 - or decrease with fewer deaths



> Predictive genetic tests

- Family history vs known genetic profile
 - *Anti-selection vs preferred underwriting*



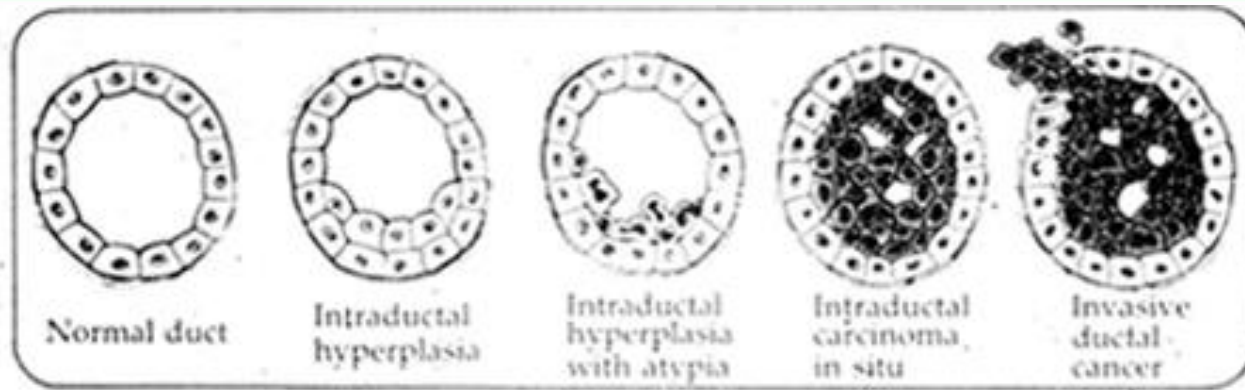
> Diagnosis history

- More screening will generate more disclosures – STP?
- Underwriting manuals need to be updated
 - Survival after pre-malignant / early cancer diagnosis
 - Cover for cancer survivors with favourable gene expressions
- Diagnostic tests involving genes (genomics) vs moratorium



**Dear Grey Area,
Why do you have to make everything
so messingly complicated.**

**Sincerely,
Fan of Black-or-White.**



- > Earlier diagnosis means **more cases where “invasion” is unclear.**
- > Problems will arise where cut-offs are based on tests that become redundant.
 - **what if Gleason score is abandoned in favour of genetic profile?**
- > **Severity requirements make claims more complicated.**
 - More evidence required
 - Evidence often unavailable in desired format
 - *Lessons from Heart Attack Troponin-T cut-offs*



- > Rapid progress in understanding of cancer

**Genes
Proteins
Biomarkers
Imaging**

**Pre-disposition
Detection
Outcome prediction
Treatment**

- > Scope for increased screening activity as over-diagnosis becomes less problematic
 - **except for Critical Illness insurers!**
- > Underwriters have development work to do
- > Genetic testing moratorium needs thorough review

Say aaaah





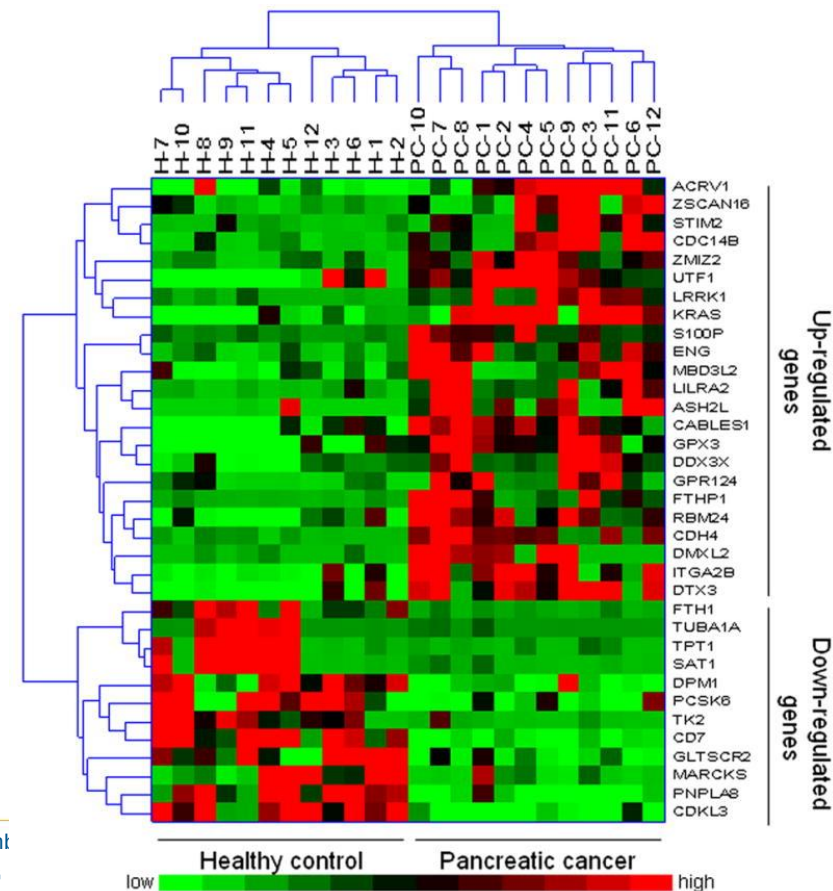
Gastroenterology. 2010 Mar;138(3):949-57.e1-7. doi: 10.1053/j.gastro.2009.11.010. Epub 2009 Nov 18.

Salivary transcriptomic biomarkers for detection of resectable pancreatic cancer.

Zhang L, Farrell JJ, Zhou H, Elashoff D, Akin D, Park NH, Chia D, Wong DT.

School of Dentistry and Dental Research Institute, University of California-Los Angeles, Los Angeles, California 90095, USA.

- Suspected biomarkers identified using gene profiling array technology
- Testing involved 30 pancreatic cancer patients, 30 chronic pancreatitis patients and 30 healthy controls
- The combination of 4 messenger RNA biomarkers differentiates pancreatic cancer patients from non-cancer subjects with 90% sensitivity and 95% specificity

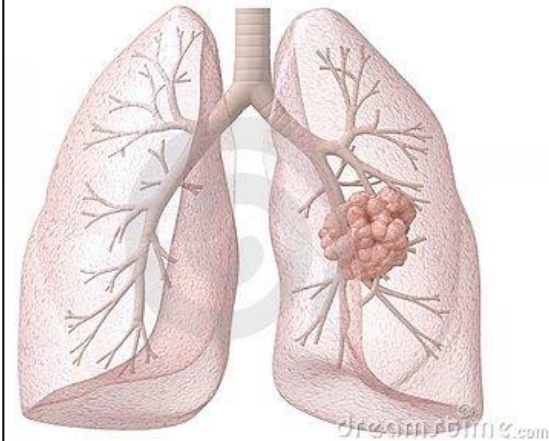




© 2012 by The American Society for Biochemistry and Molecular Biology, Inc.
This paper is available on line at <http://www.mcponline.org>

Proteomic Analysis of Human Saliva From Lung Cancer Patients Using Two-Dimensional Difference Gel Electrophoresis and Mass Spectrometry*

Hua Xiao^{‡|||}, Lei Zhang[‡], Hui Zhou[‡], Jay M. Lee^{§**}, Edward B. Garon[¶],
and David T. W. Wong^{‡||**†§§¶¶}



J Mol Med (Berl). 2012 Apr;90(4):427-34. doi: 10.1007/s00109-011-0829-0. Epub 2011 Nov 18.

Salivary transcriptomic biomarkers for detection of ovarian cancer: for serous papillary adenocarcinoma.

Lee YH, Kim JH, Zhou H, Kim BW, Wong DT.

Graduate Institute of Biomedical Engineering, National Central University, Jhongli City, Taiwan, Republic of China.

PLoS One. 2010 Dec 31;5(12):e15573. doi: 10.1371/journal.pone.0015573.

Discovery and preclinical validation of salivary transcriptomic and proteomic biomarkers for the non-invasive detection of breast cancer.

Zhang L, Xiao H, Karlan S, Zhou H, Gross J, Elashoff D, Akin D, Yan X, Chia D, Karlan B, Wong DT.

School of Dentistry and Dental Research Institute, University of California Los Angeles, Los Angeles, California, United States of America. leizhang@ucla.edu







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